



## COSC 5580 Fall 2023

### Project Assignment 1 - Report

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## Overview

The currency exchange rate has always dominated international trade and growth of a country. Both the prediction and description of exchange rate yields important information regarding a country's growth. One assumption that is often made is that historical data should be able to incorporate the different factors affecting the exchange rate. Many researchers have constructed models using such historical data and is an active area of research. For this project, we are trying to understand the different factors that are responsible for affecting the exchange rate of currencies. For predictive analysis, the data being in a time-series format we can use different forecasting methods and for descriptive analysis, we can perform dimensionality reduction among other techniques and find the relative importance of different factors.

## Background

### Currency Exchange

Currency exchange or Foreign exchange is termed as the conversion of one country's currency into another and is dependent on a lot of economic factors such as trade, geopolitical stability, etc. It also gives a metric of the demand of that currency globally, i.e. it tells you how much of one currency you need to buy one unit of the other currency.

### Previous Research

Work from Karakostas [2] presents a latest survey on the different research trends and previous research. This paper gave us the relevant background needed to understand the problem space. Kamruzzaman and Sarker's [1] work on using Artificial Neural Networks (ANN) for forecasting the currency exchange presents some interesting metrics and simulation results.

## Problem Statement

The broad goal of this project is to find the relationship between different factors that can potentially affect the currency exchange rate between different countries. For our project we have decided to base this study on the exchange rate of USD\$ to Japanese Yen. We will

try to understand the relationship between different news events and the exchange rate as a contribution of the project.

Data Science questions

1. How are factors such as Inflation, GDP per capita affect the currency exchange rate of the country?
2. What features are more strongly or weakly related to the exchange rate?
3. By looking at the historical data how volatile is the exchange rate?
4. Does the exchange rate follow any general trend with respect to different features?
5. Is there any trend visible from plotting different variables with respect to the exchange rate?

## Datasets

The primary data sources are

1. Federal Reserve Economic Data (FRED) - Using API to get data
2. Department of the Treasury and the Bureau of the Fiscal Service - Downloading from the source
3. Scraping for text news events
4. Yahoo Finance API for Stock market data

Below is the aggregate list of features we collected across all the dataset and their significance

1. Consumer Price Index - It measures the change in the price of a specific group of goods and services and is commonly used to measure the level of inflation.
2. Interest Rate - It is the fee paid to the lender or investor for borrowing or investing funds, usually expressed as a percentage, reflecting the cost or return on funds.
3. Inflation - It is the sustained increase in the general price level in an economy over a period of time, resulting in a decrease in the purchasing power of money.
4. Exchange Rate - It is the price at which one currency can be exchanged for another.
5. GDP Per Capita - It is the total economic output of a country divided by its entire population and is used to measure the level of economic production per average citizen.
6. Debt held by the Public - It is the borrowed and outstanding debt owed to others by governments, corporations, or individuals and held by investors, banks, or other creditors.
7. Intra-governmental Holdings - It is the debt which the government owes and controls a portion of a domestic enterprise or institution's equity, usually to exert influence over its operations and policy direction or to achieve public policy objectives.

8. Total Public Debt Outstanding - It is the cumulative amount of all borrowing and debt, including bonds and other forms of debt, that the Government still needs to repay.
9. Government Gross Debt - It is the total debt the government owes to various creditors and usually includes bonds, loans, and other borrowings used to finance government expenditures and programs.
10. Stock market data (S&P and Nikkei\_225) - It is the exchange of shares and stocks by people who represent ownership of various companies.

The stock market features are -

- (a) Date
  - (b) Opening price
  - (c) Closing price
  - (d) High (Highest price for the day)
  - (e) Low (Lowest price for the day)
  - (f) Volume
11. Date - Date of collecting the data
  12. News text data (From BBC data) - Scraped data from BBC news API for both USA and Japan.

Note - Some features are for both the countries - US and Japan. Therefore, we only mention it once here.

## Data Issues

For the most part, the data was sufficiently clean and usable but below are some of the concerns

- **Data Scraping sufficiency** (For text data) - The scraping data is considered based on yearly major events which we are considering as news (or events), hence our data for this will have fewer rows as compared to other data sources. Logically, these types of events influence a country's development not on a daily basis but on monthly, quarterly or more commonly yearly basis.
- **Features** - Features such as GDP and Inflation are not collected daily or sometimes not even monthly. They are reported annually or quarterly.
- **Paid data sources** - Some data sources with useful features were available but were behind a paywall.
- **Missing Data** - Presence of missing data can be seen in some datasets.

## References

- [1] KAMRUZZAMAN, J., AND SARKER, R. A. Ann-based forecasting of foreign currency exchange rates. *Neural Information Processing-Letters and Reviews* 3, 2 (2004), 49–58.
- [2] KARAKOSTAS, E. The significance of the exchange rates: A survey of the literature. *Modern Economy* 12, 11 (2021), 1628–1647.